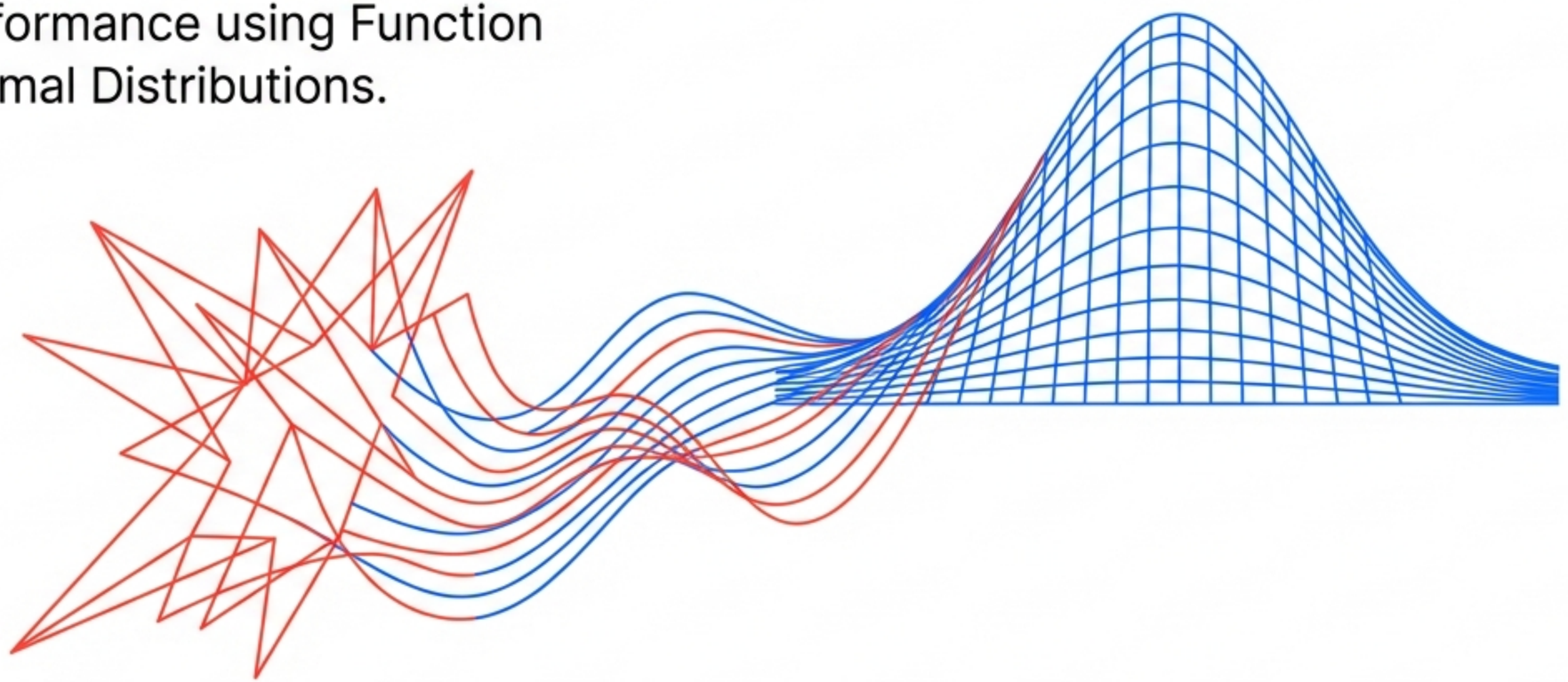
Series: Feature Engineering / Chapter 4

Mathematical Transforms in Feature Engineering

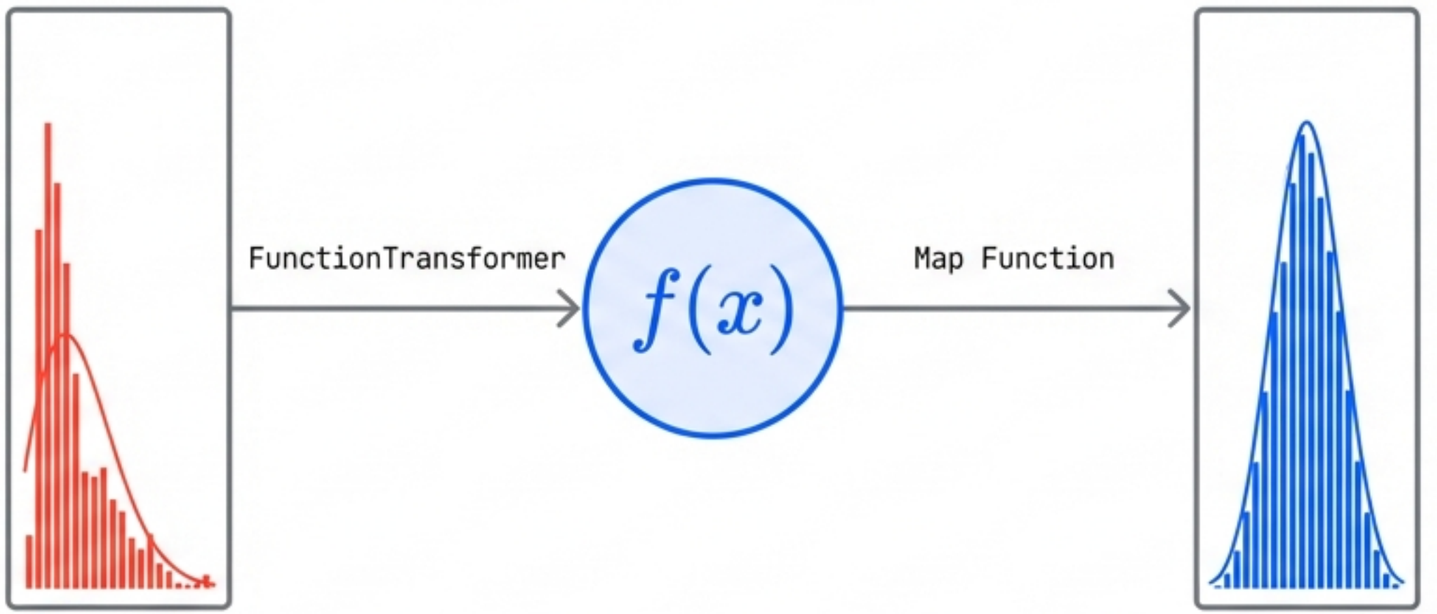
Optimising Model Performance using Function Transformers and Normal Distributions.



Concept: Transformation. Goal: Normality. Outcome: Accuracy.

Reshaping the Probability Distribution

Feature Engineering extends beyond encoding and imputation. It involves altering the fundamental geometry of data to assist algorithmic learning.

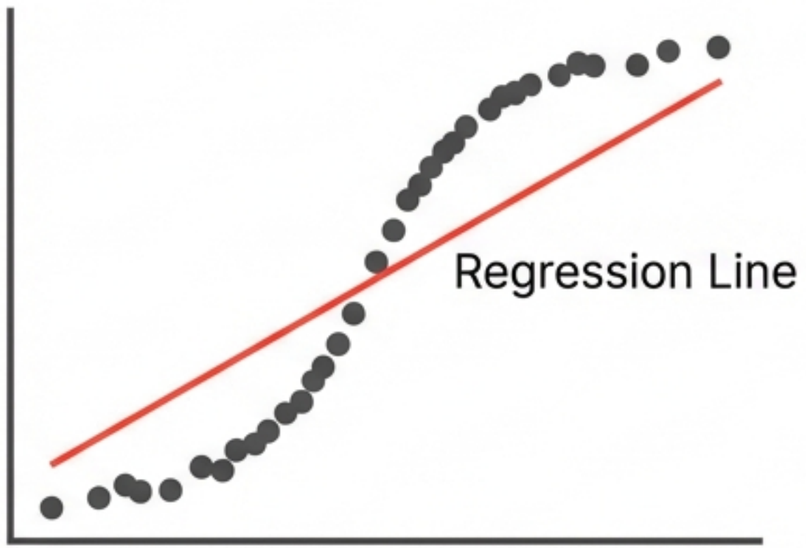


The Toolbox

- Log Transform (Primary Focus)
- Reciprocal Transform
- Power Transforms (Square & Square Root)

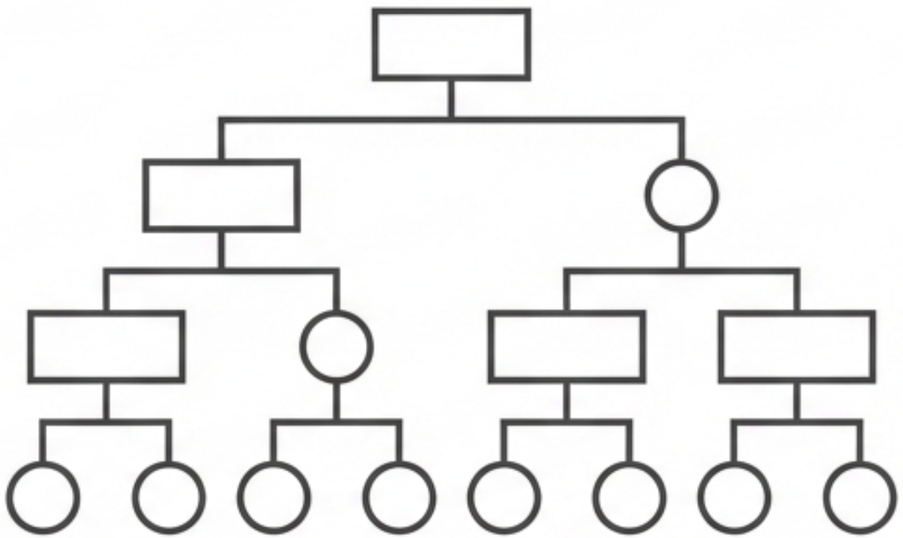
The Gaustian Obsession: Why Normality Matters

Linear Models
(e.g., Logistic Regression)



Assumes Normality. Performance degrades on skewed data.

Tree Models
(e.g., Decision Trees)



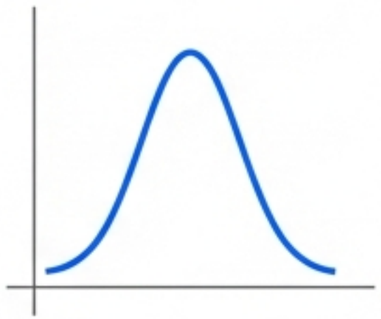
Distribution Agnostic. Splits based on value thresholds, not probability curves.

Statistics “loves” the Normal Distribution.
It is the happy place where math becomes simple.

Diagnosing the Distribution

The Diagnostic Toolkit

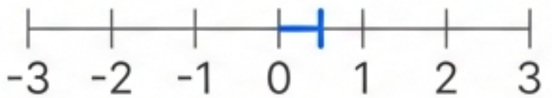
PDF (Probability Density Function)



Visual Inspection via `sns.distplot`

◉ Subjective

Skewness Score

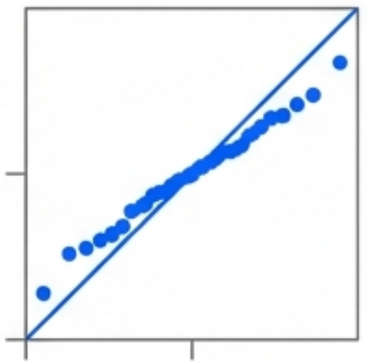


Calculation with `pandas.skew()`

0 = Normal. +/- = Skewed.

◉ Quantitative

Q-Q Plot

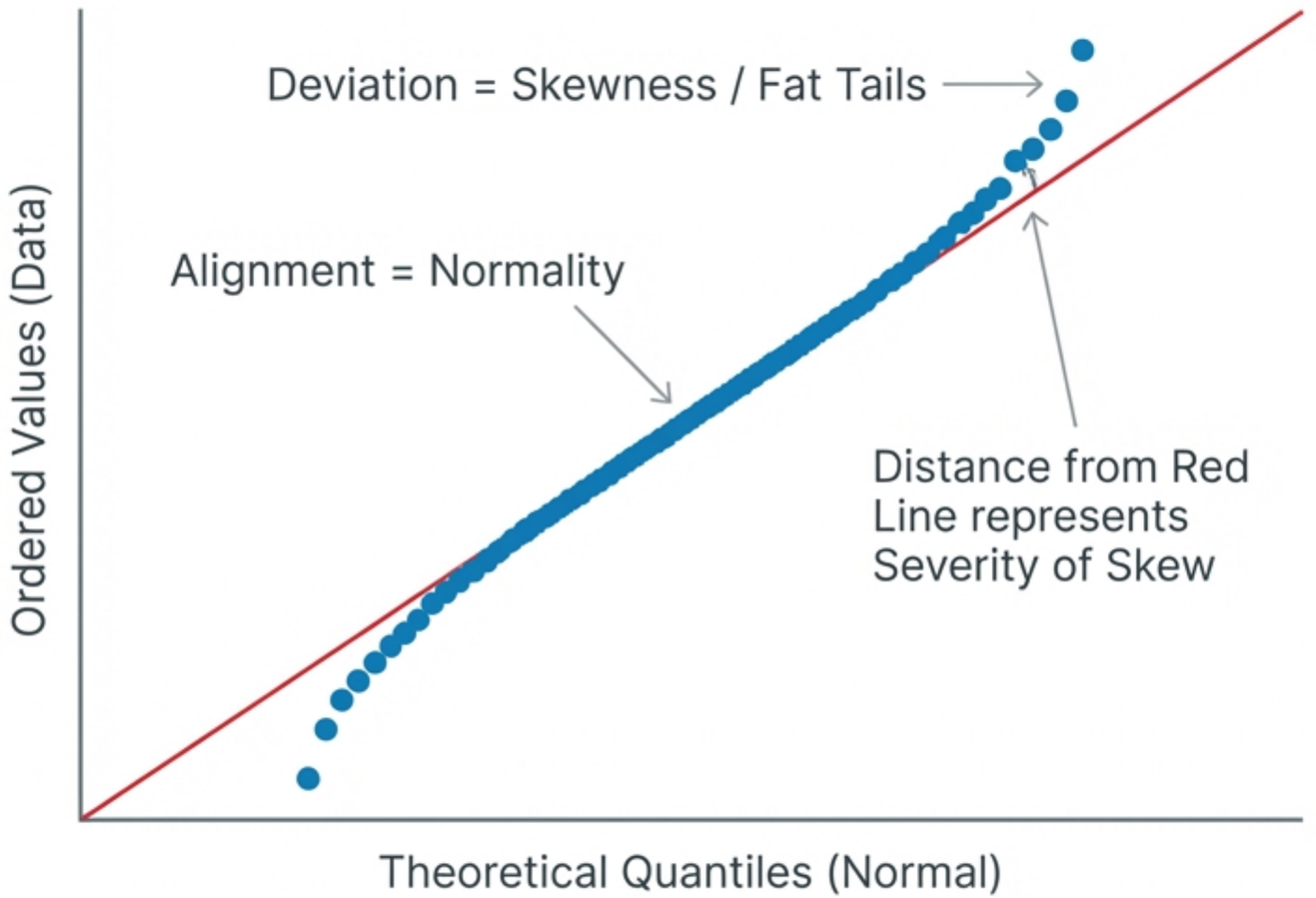


Quantile-Quantile Plot

The Gold Standard for diagnosis.

◉ Definitive

Visual Literacy: Reading the Q-Q Plot



Rule of Thumb: If the blue dots hug the red line, the patient is healthy (Normal). If they peel away, treatment is required.

The Treatment Tool: Scikit-Learn FunctionTransformer

untitled.py

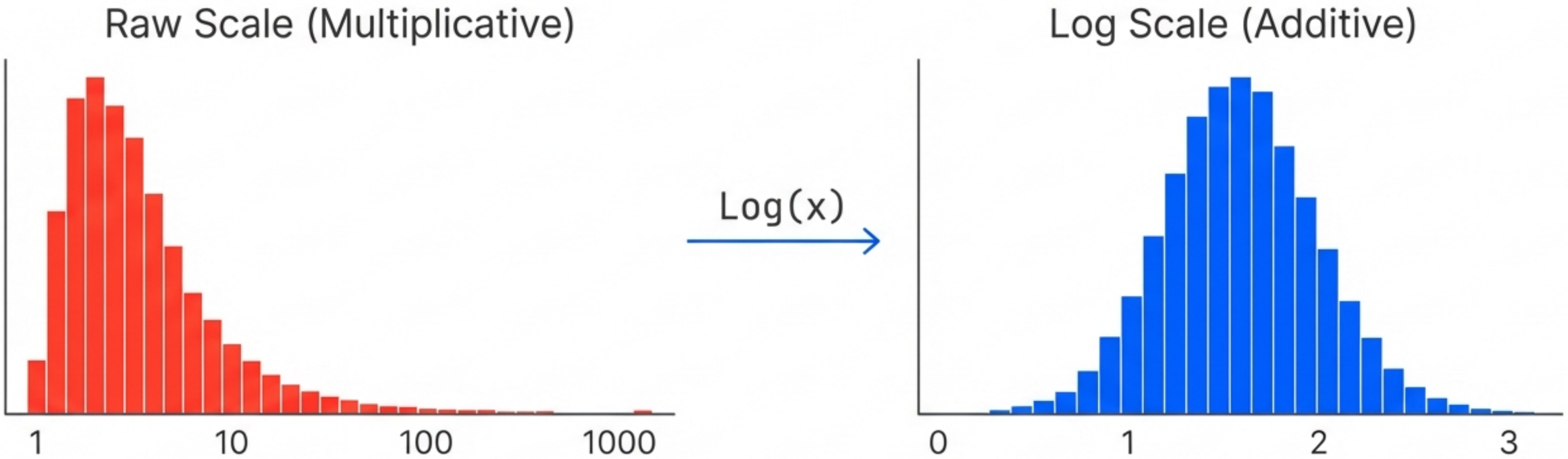
```
from sklearn.preprocessing import FunctionTransformer
import numpy as np

# 1. Define the Transformer (e.g., Log Transform)
# handling zeros with log1p (log(1+x))
trf = FunctionTransformer(func=np.log1p)
trf = FunctionTransformer(func=np.log1p)

# 2. Apply to Data
X_transformed = trf.fit_transform(X)
```

Flexibility & Safety

- **Flexibility**: Maps any mathematical function to a feature.
- **Safety**: Use `np.log1p` instead of `np.log` to prevent errors with zero values.

Treatment A: The Log Transform

The Gold Standard for Right-Skewed Data



Mechanism: Compresses the scale of large values. It pulls the long tail back toward the center.

Contraindication: Cannot be used on negative values.

Alternative Treatments

Reciprocal Transform

$$1/x$$

Reverses the order. Large becomes small, small becomes large.

Note: Unstable with zero values. Inter

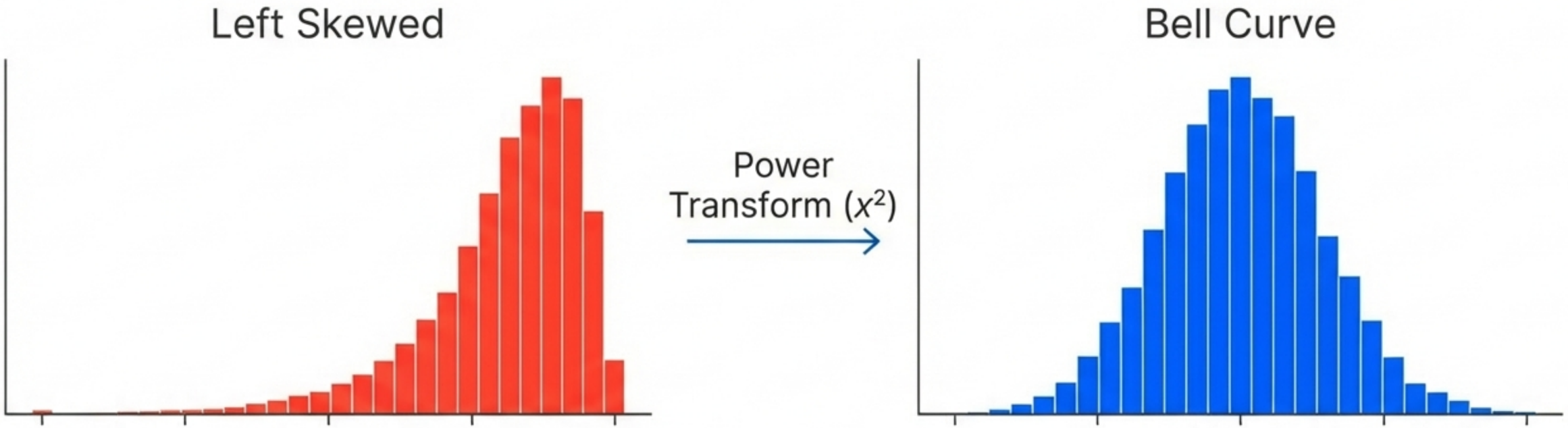
Square Root Transform

$$\sqrt{x}$$

A milder version of Log. Good for right-skewed data where Log is too aggressive.

Treatment C: The Square Transform

Correcting Left-Skewed Data

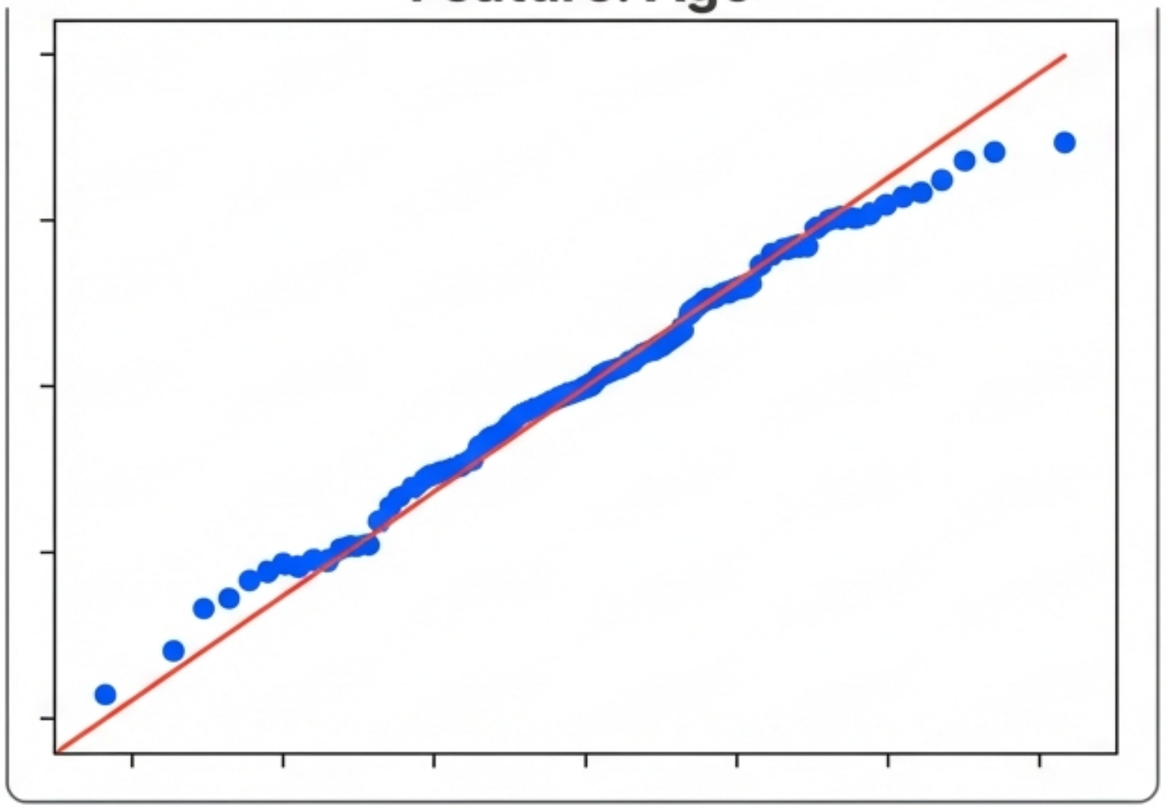


Mechanism: Expands the distance between smaller values, countering the bunching at the high end.

Case Study: Titanic Dataset

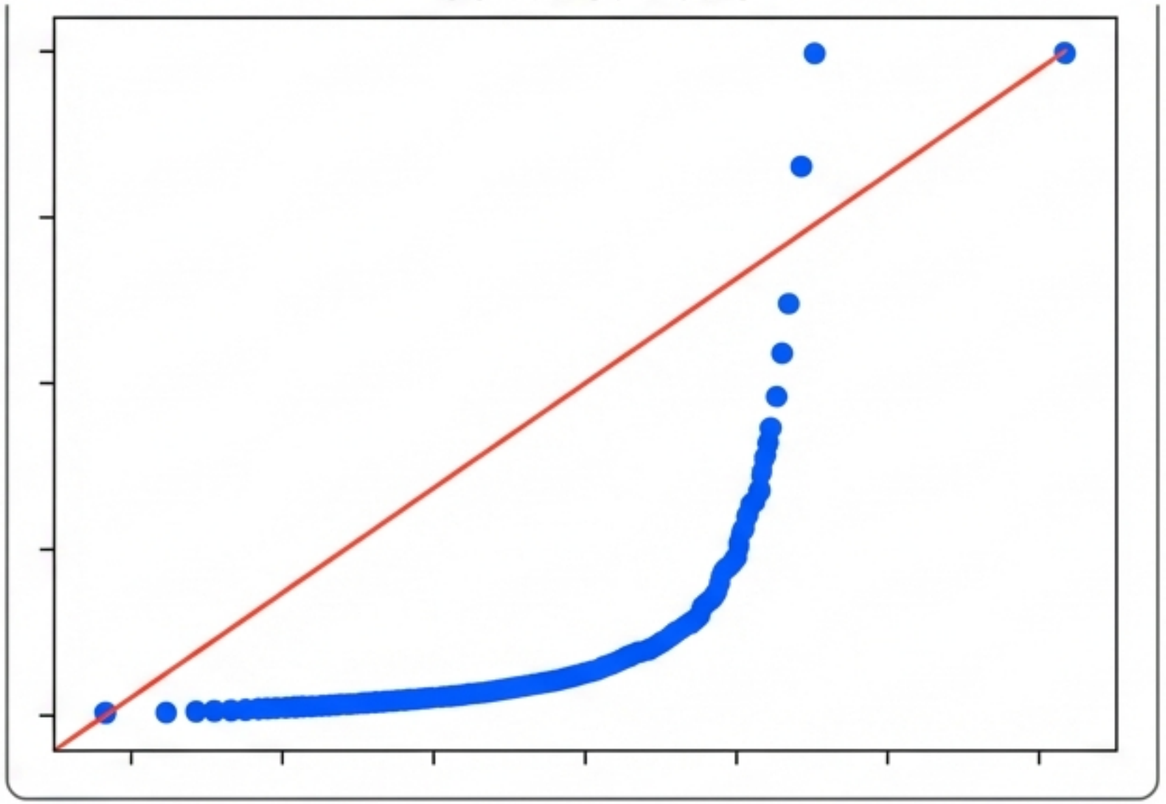
Initial Diagnosis of Features

Feature: Age



Status: Roughly Normal

Feature: Fare

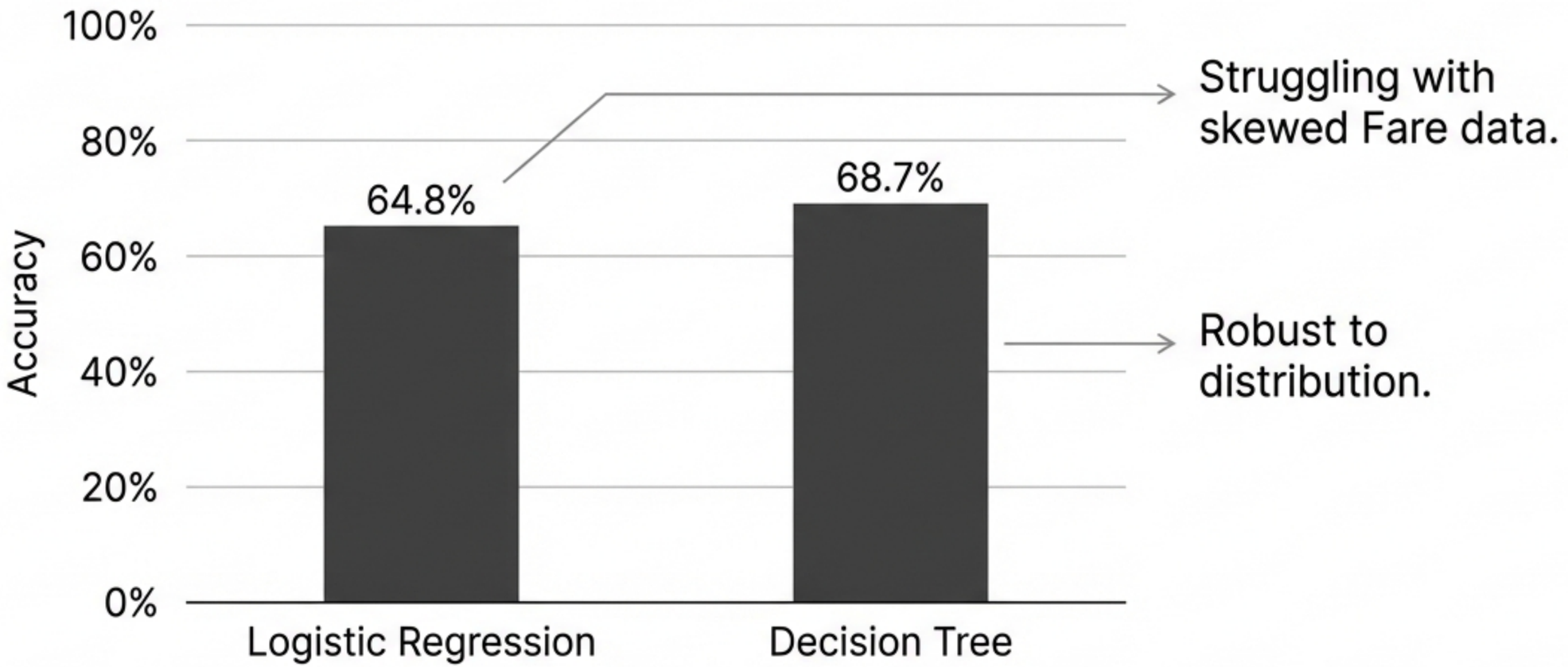


Status: Critical Right Skew (Outliers)

Diagnosis: "Age" is healthy. "Fare" exhibits severe non-normality due to extreme ticket prices.

Experiment Phase 1: Baseline Performance

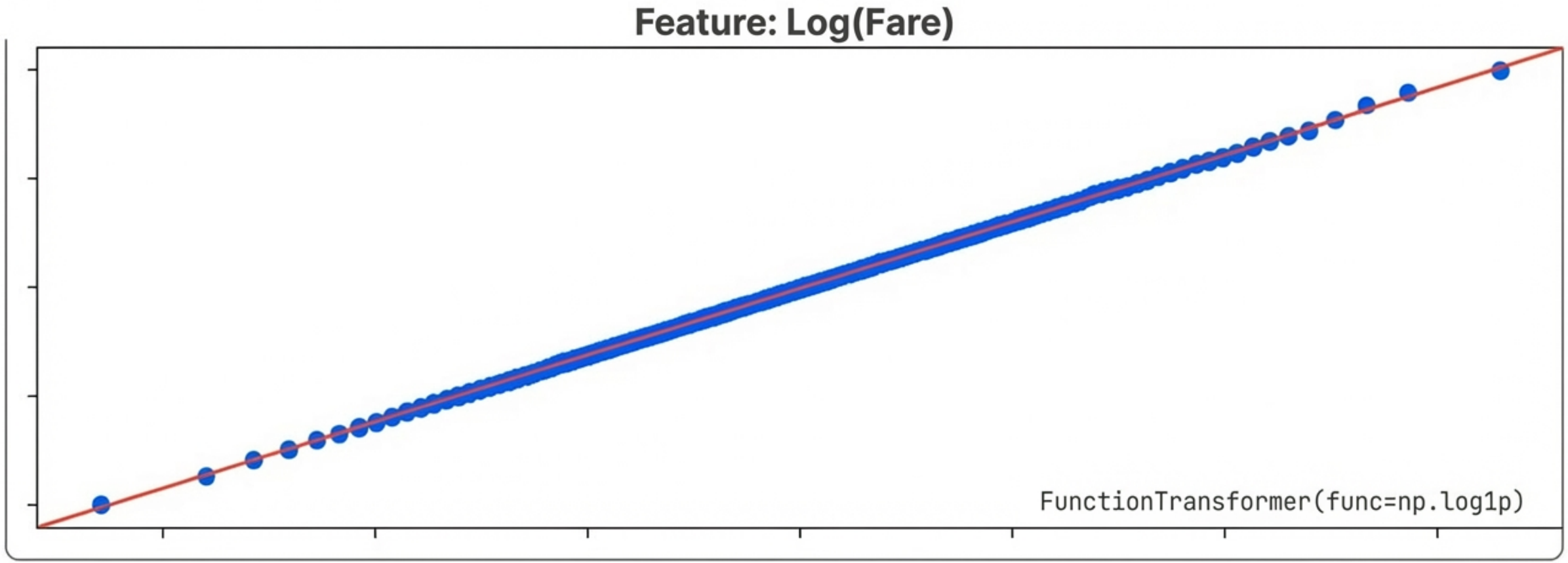
Training on Raw, Untransformed Data



Observation: The Linear model is significantly underperforming the Tree model.

Applying the Cure

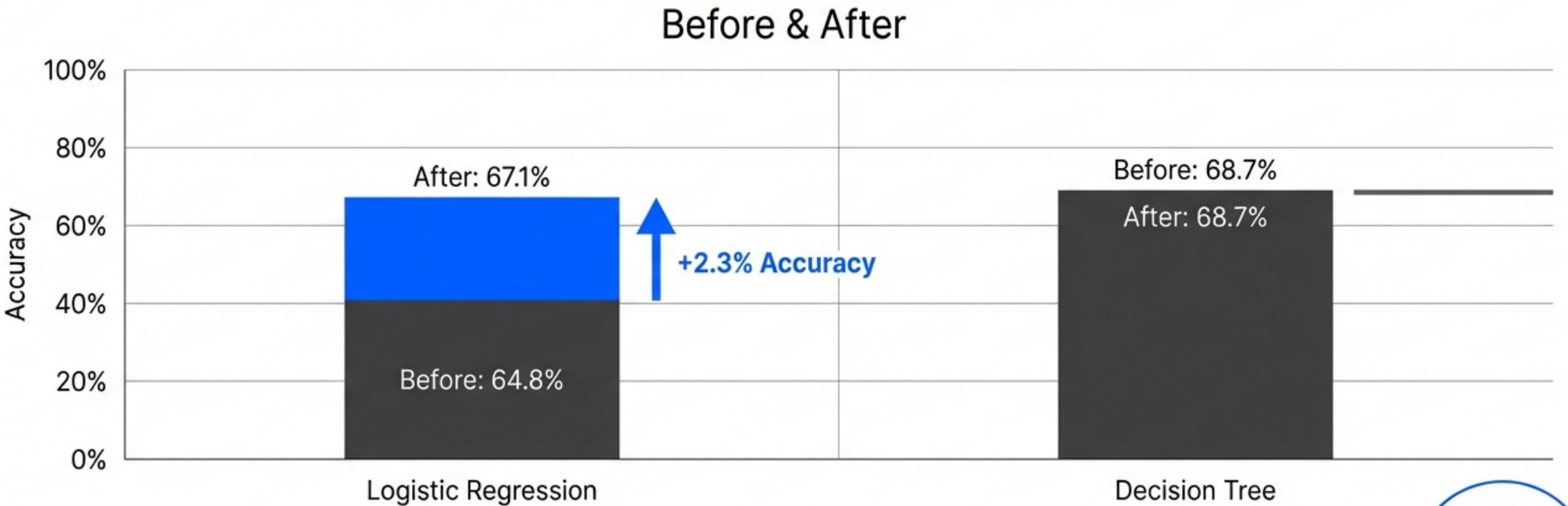
Intervention: Log Transform on ‘Fare’ column



Result: The distribution has successfully mimicked Normality.

The Post-Treatment Prognosis

Performance Uplift Validation

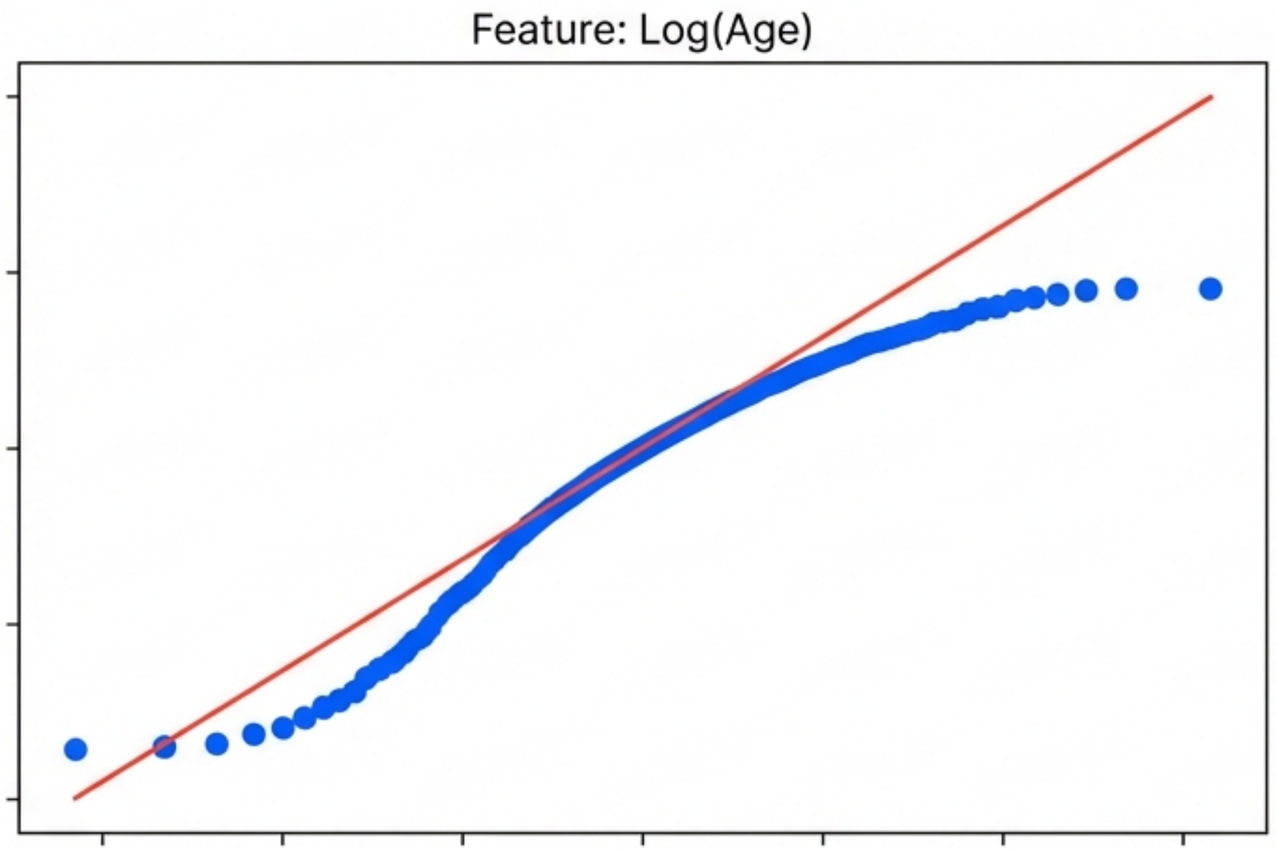


Conclusion: Mathematical transforms unlock the hidden potential of Linear Models. Tree models remain unaffected.

Validated via 10-Fold Cross-Validation

Caution: Do Not Over-Prescribe

The Risk of Treating Healthy Features



Action: Applied Log Transform to “Age” (Already Normal).

Outcome: Model Accuracy Decreased.

Lesson: Only treat features that are sick. Verify with a Q-Q plot post-transform.

Practical Heuristics in Helvetica Now Display bold

Right Skewed?

log **Action:** →
Try Log Transform.

Fallback: If Log is too strong, try Square Root.

Safety: Use `log1p` for zeros.

Left Skewed?

x^2 **Action:** →
Try Square (x^2) or Exponential.

Validation Protocol

Action:
1. Check Q-Q Plot.
2. Check Accuracy Metric.

“If the plot looks better but accuracy drops, trust the accuracy.”

Implementation: `sklearn.preprocessing.FunctionTransformer`